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Perspective

A conceptual framework for understanding phase separation and addressing open questions and challenges

Tanja Mittag[1,*] and Rohit V. Pappu[2,*]
[1]Department of Structural Biology, St. Jude Children's Research Hospital, 262 Danny Thomas Place, Memphis, TN 38105, USA
[2]Department of Biomedical Engineering and Center for Science & Engineering of Living Systems (CSELS), Washington University in St. Louis, St. Louis, MO 63130, USA
*Correspondence: tanja.mittag@stjude.org (T.M.), pappu@wustl.edu (R.V.P.)
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SUMMARY

Macromolecular phase separation is being recognized for its potential importance and relevance as a driver of spatial organization within cells. Here, we describe a framework based on synergies between networking (percolation or gelation) and density (phase separation) transitions. Accordingly, the phase transitions in question are referred to as phase separation coupled to percolation (PSCP). The condensates that result from PSCP are viscoelastic network fluids. Such systems have sequence-, composition-, and topology-specific internal network structures that give rise to time-dependent interplays between viscous and elastic properties. Unlike pure phase separation, the process of PSCP gives rise to sequence-, chemistry-, and structure-specific distributions of clusters that can form at concentrations that lie well below the threshold concentration for phase separation. PSCP, influenced by specific versus solubility-determining interactions, also provides a bridge between different observations and helps answer questions and address challenges that have arisen regarding the role of macromolecular phase separation in biology.

INTRODUCTION

Membraneless biomolecular condensates are thought to provide spatial and temporal organization of cellular matter (Banani et al., 2017). There are numerous condensates in cells and the simplest operational definition of a membraneless biomolecular condensate is *of an entity that is not bound by a membrane that concentrates specific types of biomolecules* (Banani et al., 2017). To this, we add the rider that condensates are most likely non-stoichiometric assemblies of multiple proteins and nucleic acids (Choi et al., 2020a). Beyond these prescriptions, the name does not commit to a process by which condensates form or dissolve nor does it commit to specific material properties or structure-function relationships. However, as the term implies, condensation, a proxy term for macromolecular phase separation, is likely to be an important process by which condensates form and dissolve (Alberti et al., 2019). The focus on macromolecular phase separation builds on some of the earliest suggestions that invoked this process as a way to enable spatial organization of cellular matter (Walter and Brooks, 1995; Wilson, 1899). Here, we aspire for precision in defining the concept of phase separation, generalizing the concept to focus on the resonance rather than perceived dissonance with classic structure-function paradigms.

WHAT IS PHASE SEPARATION?

The simplest example of phase separation is that of a liquid-vapor transition for a one-component fluid (Rowlinson, 1979). For certain combinations of pressure and temperature, the fluid separates into two coexisting phases namely a dilute phase known as the vapor that coexists with a dense phase known as the liquid. Here, a phase is defined by uniformity of the average density of the fluid within the volume that is taken up by the phase in question. If ρ_L and ρ_V are the average densities of the coexisting liquid and vapor phases, respectively, then the total density for the combination of pressure and temperature where the fluid separates into two coexisting phases is written as $\rho = (\varphi_L\rho_L + \varphi_V\rho_V)$. Here, φ_L and φ_V are the volume fractions of the system taken up by the liquid and vapor phases, respectively. The densities of the coexisting phases are set by joint requirements of chemical, thermal, and mechanical equilibria among the coexisting phases. This means that chemical potentials are equalized across the phase boundary, as are the temperatures and pressures.

For a simple one-component fluid, it is easy to appreciate phase separation as a density transition (Rowlinson, 1979). Importantly, this definition holds in multicomponent systems as well. To illustrate this point, we consider a ternary mixture in a closed system comprising macromolecules A and B in a solvent S. A classic example of such a mixture is of polyethylene glycol (macromolecule A), dextran (macromolecule B), and water (the solvent S) (Münchow et al., 2008). These systems undergo "aqueous two-phase separation." In aqueous two-phase separation, the interactions of A and B are incompatible with one another. However, both macromolecules are defined by favorable interactions with the solvent S. In such a scenario, the

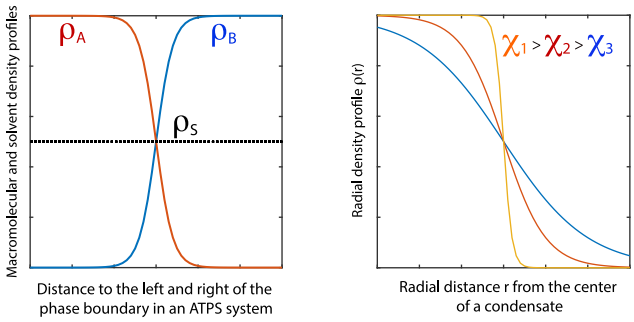

Figure 1. Phase separation is a density transition
(Left) In an aqueous two-phase system (ATPS), the interactions of A and B are incompatible with one another, but both macromolecules make favorable interactions with the solvent S. The mixture separates into two coexisting phases, one that is rich in A, the other rich in B. The density of B (ρ_B) in the A-rich phase is low and the density of A (ρ_A) in the B-rich phase is low. In this case, the solvent density is the same across the two phases, but in general, it is governed by the ratio of A:B, the volumes taken up by the A- and B-rich phases, and the extent to which it prefers to interact with A versus B. (Right) For a binary mixture comprising a macromolecule and solvent, the radial density profile from the center of a condensate shows a sharp, χ-dependent decrease in macromolecular density as the phase boundary is crossed.

mixture separates into two coexisting phases, one that is rich in A and the other rich in B. The density of B in the A-rich phase will be low, and likewise, the density of A in the B-rich phase will be low (Figure 1A). The solvent density will be governed by the ratio of A:B, the volumes taken up by the A- and B-rich phases, and the extent to which it prefers to interact with A versus B. Figure 1A depicts the case where the solvent density is the same across the two phases.

Therefore, in a system with *n* components, the density is a multicomponent vector denoted as $[\rho_1, \rho_2, \ldots, \rho_n]$. Two or more coexisting phases can arise when one, all, or a small subset of the components undergo a density transition, also known as phase separation. This type of phase behavior arises from some form of incompatibility or unfavorable interactions among a subset of the components. It can also arise from the complexation of oppositely charged molecules, as is the case with phase separation via complex coacervation (Pak et al., 2016), which also involves repulsive interactions among like-charged species, charge regulation effects (Fossat et al., 2021), and the partitioning into or release of solution ions from complexes or dense phases (Heidarsson et al., 2022; Sing and Perry, 2020).

Based on this discussion, there is a clear prescription for deciding if a condensate (following the rather loose definition of Banani et al.) arises via phase separation. On all length scales, are the concentration ratios or relative densities of the macromolecules and solvent uniform and in accord with the bulk values of these ratios? If the answer is yes, then we have a stable, well-mixed, one-phase system (Flory, 1942; Huggins, 1941). However, if there are density inhomogeneities on length scales that are larger than macromolecular dimensions and on a par with the dimensions of the volume of the system of interest, then a necessary case exists for a density transition. Whether phase separation is the operative mechanism and whether the high local concentration of specific collections of macromolecules, defined as a condensate, is a distinct phase will require answering the following questions (Figure 1B): (1) are the average densities of at least some of the constituent macromolecules within a condensate different within the condensate when compared with the surrounding milieu? And (2), are the fluctuations around the average densities within and outside the condensate, which result from dynamic exchange of components, on a par with the fluctuations we expect based on the molecular dimensions? If the answers to these questions are affirmative, then the condensate and the non-condensate regions may be viewed as distinct, coexisting phases that arise through some form of macromolecular phase separation. These discussions lead us to the clearest signatures of macromolecular phase separation *viz*., the existence of a *saturation concentration*.

DEFINING SATURATION CONCENTRATION IN SIMPLE BINARY MIXTURES

To define the concept of a saturation concentration, we shall consider a simple binary mixture that includes a macromolecule in a solvent. The relevant energy scale is defined by a dimensionless parameter χ, which quantifies the difference between the strengths of macromolecule-solvent interactions and the sum of inter-macromolecule and solvent-solvent interactions (Flory, 1942; Hildebrand, 1981; Huggins, 1941).

If χ is negative, macromolecule-solvent interactions are favored, on balance, and the expectation is that the binary mixture always forms a well-mixed, one-phase system. If $\chi = 0$, then the macromolecule-solvent interactions are perfectly counterbalanced by the sum of inter-macromolecule and solvent-solvent interactions. This would be an ideal mixture, defined purely by the entropy of mixing. If χ is positive, then, on balance, macromolecule-solvent interactions are unfavorable. In this scenario, for a fixed χ, i.e., for a fixed set of solution conditions, there will be a threshold macromolecular concentration above which the well-mixed, one-phase regime is saturated. We designate this threshold concentration as the saturation concentration or c_{sat}. Above c_{sat}, the free energy of mixing is unfavorable, and thermodynamic stability is achieved by separation of the mixture into coexisting phases *viz*., a macromolecule-poor dilute phase and a macromolecule-rich dense phase. A phase boundary is set up and both macromolecular chemical potentials and the osmotic pressures are equalized across the phase boundary (Zeng et al., 2020). This equalization will determine the equilibrium concentration of the macromolecule in the dilute phase namely c_{sat} and the macromolecular concentration in the dense phase namely c_{dense}. Two different macromolecules that have the same χ parameter will have similar saturation concentrations. However, this does not mean that the macromolecules must have identical or even similar biochemical functions nor does it mean that the dense phases formed by the macromolecules have similar material properties or functions.

For a given value of χ and all values of the bulk concentration that are greater than c_{sat}, the macromolecular concentration in the dilute and dense phases should be c_{sat} and c_{dense}, respectively. The existence of a saturation concentration, which is a necessary condition for phase separation, has been established for various binary mixtures *in vitro* (Brady et al., 2017; Bremer et al., 2022; Crick et al., 2013; Elbaum-Garfinkle et al., 2015;

Guillén-Boixet et al., 2020; Kar et al., 2022; Martin et al., 2020; Molliex et al., 2015; Nott et al., 2015; Wang et al., 2018; Wei et al., 2017; Yang et al., 2020). It has also been established in cells for specific types of macromolecules whose expression levels are controlled using specific approaches. In cellular experiments, the concentration of a macromolecule of interest is titrated either by controlling expression levels or by modulating the interaction energies using optogenetic tools (Alberti et al., 2018; Bracha et al., 2018; Freibaum et al., 2021; Fritsch et al., 2021; Guillén-Boixet et al., 2020; Klosin et al., 2020; Shimobayashi et al., 2021; Shin et al., 2017; Yang et al., 2020).

MULTIVALENT MACROMOLECULES ALSO UNDERGO NETWORKING (PERCOLATION) TRANSITIONS

In addition to their solubilities, which vary with solution conditions, biomacromolecules are known for exquisite specificity of biochemical interactions. This gives rise to specificity of molecular recognition via binding. A full description of binding requires measurements that help with constructing binding polynomials (Wyman and Gill, 1990). It is worth noting that the definition of χ does not feature in formal descriptions of binding polynomials. This is because binding and phase equilibria are separable albeit linked thermodynamic processes. As a result, macromolecules that have the same solubility profiles can have very different binding specificities.

In addition to solubility-determining interactions, biomacromolecules can engage in site-specific interactions if they involve macromolecules of well-defined three-dimensional structures. Alternatively, they can engage in chemistry- or sequence-specific interactions if the associations involve conformationally heterogeneous systems such as intrinsically disordered proteins. The specific interaction motifs are either hotspots on folded domains, specific linear motifs in disordered regions, or even individual residues. Collectively, the specific interaction motifs are known as *stickers*, and the reversible associations between stickers can be thought of as physical crosslinks. Each of the physical crosslinks can be hydrogen bonds, ionic interactions, cation-π, or π-π interactions, or associations among hydrophobic groups. Physical crosslinks between stickers are precisely the types of interactions that give rise to specificity in the form of sequence-specific folds (Eisenberg, 2003), fold-specific molecular recognition (Alexov and Honig, 2010), and assemblies of precise stoichiometries (Khayat et al., 2005).

Macromolecules that drive condensate formation feature either intrinsic multivalence or emergent multivalence of stickers, the latter achieved by oligomerization or polymerization (Choi et al., 2020a; Marzahn et al., 2016). Multivalence of stickers and the network of reversible associations can become generators of molecular clusters with an assortment of stoichiometries. These stoichiometries are governed by the strengths of inter-sticker crosslinks, the valence of stickers, and the topologies of the molecules that form intra- and inter-molecular inter-sticker crosslinks. In general, if there are more than three stickers per molecule and inter-sticker interactions form mainly between and not within molecules, then the clusters that form via networks of physical crosslinks will grow in both size and number as concentrations increase (Choi et al., 2020a, 2020b). Importantly, there will be a valence- and topology-specific threshold concentration known as the percolation threshold or c_{perc} above which the largest cluster becomes system spanning. This type of system-spanning or percolated network is also known as a physical macrogel. Physically crosslinked networks that form via percolation transitions, also known as sol-gel transitions, have distinctive network structures (Cohan and Pappu, 2020; Harmon et al., 2017). Furthermore, the sol that is prevalent below c_{perc} is made up of an assortment of clusters of finite sizes and stoichiometries. The size distributions of clusters that form in the sol have specific functional forms that are governed by the valence of stickers, the topologies of the macromolecules (Bhandari et al., 2021; Choi et al., 2019, 2020a; Schmit et al., 2020, 2021), and the interaction strengths of stickers. Unlike phase separation, the sol-gel transition also known as the percolation transition is continuous, and well above c_{perc}, essentially all molecules are incorporated into the percolated network.

PERCOLATION AND PHASE SEPARATION CAN BE COUPLED TO ONE ANOTHER

Percolation is a *networking transition* enabled by the multivalence of specific interactions. In contrast, phase separation is a *density transition* enabled by the totality of interactions that contribute to χ. Formally, the measured values of χ combine two contributions. The first, influenced by overall composition, is the main determinant of solubility. The second corresponds to the contributions of specific interactions. Tanaka has proposed that the phase behavior of associative polymers is governed by a renormalized χ written as: $\chi' = \chi + \Delta\chi$. Here, $\Delta\chi$ accounts for the specific inter-sticker interactions, which introduces a concentration dependence to the original χ parameter (Tanaka, 2006). Note that the renormalization of χ can also occur through specific interactions with ligands (Ruff et al., 2021a) and/or through sequence-specific interactions with macromolecular surfaces (Morin et al., 2022).

Phase transitions realized by the renormalized χ, referred to as χ', will come in two forms (Figure 2). If c_{perc} is less than c_{sat}, then *percolation occurs without phase separation*. In this scenario, the question is if the total macromolecular concentration in a binary mixture is less than or greater than c_{perc}. If it is the former, then the system, which is a sol, is described by a collection of finite-sized, pre-percolation clusters. If the total macromolecular concentration is greater than c_{perc}, then a physical gel, known as a macrogel or system-spanning network forms, and this network will span the entire volume available to the system. If, on the other hand, the total macromolecular concentration in a binary mixture is greater than c_{sat} *and* $c_{sat} < c_{perc} < c_{dense}$, then phase separation and percolation are coupled. Phase separation results in the formation of a condensate, and a percolated network spans the volume of the condensate, implying that the condensate is best described as a physical microgel (Vilgis, 1989). These precise descriptions of physical macrogels and microgels highlight the fact that it is incorrect to describe time-dependent changes such as "maturation" of condensates as "gelation." Instead, all condensates are physical microgels and what is referred to as maturation refers to the fact that the material

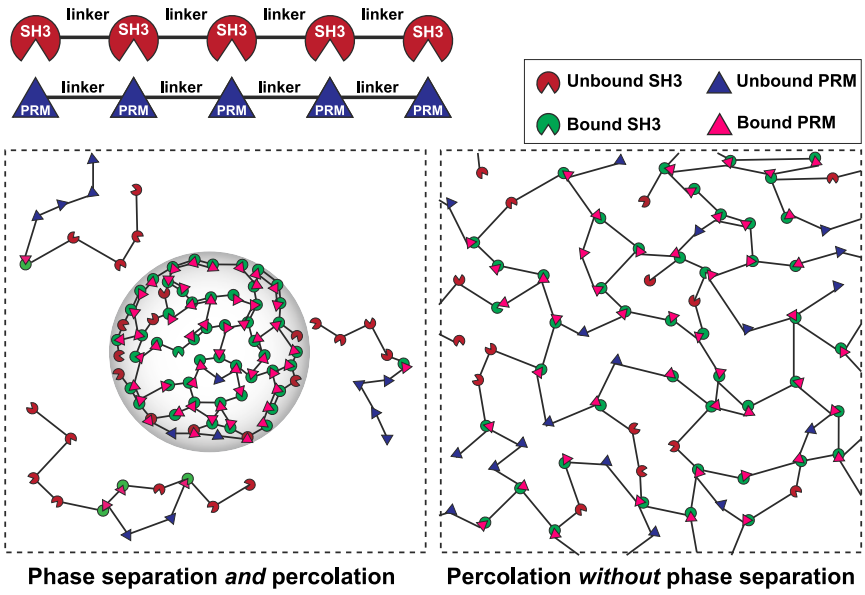

Figure 2. Coupling and decoupling of phase separation and percolation in multivalent proteins

Here, we consider a system in which two proteins interact via repeats of interaction domains and motifs, e.g., via repeats of SH3 domains in protein A and repeats of proline-rich motifs (PRMs) in protein B (top); hotspots on SH3 domains and PRMs are the stickers. The multivalent interactions mediate percolation above the percolation threshold, c_{perc}. The solubility of individual protein molecules and complexes, which are strongly influenced by the linker compositions, determine whether percolation is coupled to phase separation ($c_{sat} < c_{perc}$, on the left) or proceeds without phase separation (if $c_{perc} < c_{sat}$, on the right). Clusters coexist in the dilute phase. The figure was adapted from the work of Harmon et al. (2017).

properties of viscoelastic materials are time dependent as discussed next.

CONDENSATES THAT FORM VIA PSCP ARE VISCOELASTIC NETWORK FLUIDS

The framework of phase separation coupled to percolation (PSCP) is relevant for describing the phase behaviors of multivalent macromolecules that have sticker-and-spacer architectures. Stickers are cohesive motifs that form specific, physical crosslinks, and spacers modulate the extent of coupling between phase separation and percolation. An appropriate balance of sticker valence, sticker patterning, and spacer features determines the types of PSCP transitions of associative polymers. Systems featuring stickers and spacers can be disordered linear polymers, or hybrid systems featuring folded domains and disordered regions in linear or branched architectures (Choi et al., 2020a). Rigid folded domains with sticker-and-spacer architectures belong to the class of molecules known as patchy colloids (Choi et al., 2020a).

The process of PSCP (Seim et al., 2022) leads to condensates that are viscoelastic in nature. As fluids, viscoelastic materials are network fluids, and as solids, they are elastic materials. Importantly, viscoelastic materials resist classification as equilibrium liquids or solids. Instead, their material properties are time dependent. The timescales on which viscous versus elastic properties dominate in condensates will depend on the lifetimes of the microscopic interactions mediating condensate formation, the density of physical crosslinks, and on the timescale of perturbations to the condensate. Without quantitative assessments of the stress-strain relationships as a function of time, one cannot know if the specific set of biological functions ascribed to a condensate at specific time points is the result of their properties as soft viscous fluids, hard elastic materials, or both.

Recent work has resulted in progress in experimental approaches and theoretical descriptions of condensates as network fluids, and these advances have the potential to explain why condensates are so pervasive in biology (Alshareedah et al., 2021a, 2021b; Ghosh et al., 2021; Jawerth et al., 2020; Kaur et al., 2021; Roberts et al., 2018; Shen et al., 2020; Zhou, 2021).

This much is clear: the functions of condensates, prevalent on the mesoscale (defined as the length scale encompassing hundreds to thousands of macromolecules), will be governed by the structures and dynamics of the networks of physical crosslinks formed by the network of stickers. These functions will also be determined by the overall density and the inhomogeneous spatial organization of macromolecules within condensates. Finally, the functions of condensates will also be influenced by the properties of interfaces between the condensate and the coexisting dilute phase (Folkmann et al., 2021; Figure 3).

Overall, what we know and appreciate as structure-function relationships on the molecular scale will need to be generalized, whereby the functions of condensates are described in terms of network structures, interfacial properties, and the viscoelastic moduli that are governed by the extent of physical crosslinking as well as the timescales associated with making and breaking of crosslinks. It is also worth noting that at macromolecular concentrations that are just above c_{sat}, more than 90% of the macromolecules will be in the dilute phase. This implies that the structures and size distributions in the coexisting sol fraction are also likely to play an important role in determining biochemical functions.

INTERACTIONS THAT DRIVE PHASE SEPARATION VERSUS PERCOLATION CAN BE SEPARABLE FROM ONE ANOTHER

In associative polymers, stickers engage in specific physical crosslinks, whereas spacers impact the overall solubility. For example, in synthetic systems such as ionomers (Lantman et al., 1989), less than 20% of the linear polymers are made of charged moieties, and the stickers of opposite charge make

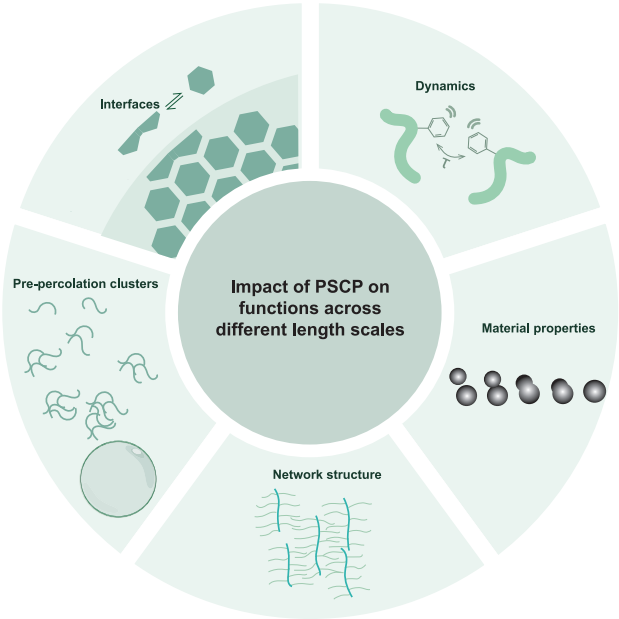

Figure 3. Consequences of condensates being viscoelastic network fluids
The properties of the multivalent macromolecules that form condensates through PSCP determine network structure and dynamics, the material properties, the properties of interfaces, and the size distributions of pre-percolation clusters.

complementary electrostatic interactions with one another. The remainder of the chain provides a scaffold that tethers the ionic residues together, and their interactions are described by generic van der Waals interactions. Therefore, stickers engage in specific physical crosslinks, whereas spacers engage in solubility-determining, non-specific interactions (Rubinstein and Dobrynin, 1997; Semenov and Rubinstein, 1998; Tanaka, 2006, 2011). Stickers, specifically their valence and interaction strengths, contribute to determining the intrinsic c_{perc} (Flory, 1941; Stockmayer, 1943). Spacers determine macromolecular solubility, which refers to the sign and magnitude of the unrenormalized or intrinsic χ (Harmon et al., 2017).

In domain-linker systems studied by Rosen and coworkers (Banani et al., 2016; Banjade et al., 2015; Li et al., 2012), the delineation of stickers versus spacers is relatively unambiguous. The extent of coupling between phase separation and percolation can be modulated through alterations of spacers or through systematic mutations to stickers (Lasker et al., 2021). This becomes more challenging for intrinsically disordered proteins. However, several insights have emerged through the deployment of combinations of biophysical methods that make it possible to quantify the extent of separation or coupling between sticker- and spacer-mediated interactions (Brady et al., 2017; Bremer et al., 2022; Martin et al., 2020; Nott et al., 2015; Wang et al., 2018). For binary mixtures involving a macromolecule and solvent, favorable interactions with the solvent lead to percolation without phase separation. This can result from the high net charge and the steric bulk of sidechains that together contribute to the effective solvation volume (Bremer et al., 2022).

A recent study (Kar et al., 2022) has demonstrated an important consequence of the separability versus coupling of specific versus solubility-determining interactions. In a pure phase separation process, subsaturated solutions, defined by bulk concentrations below c_{sat}, will be made up of dispersed monomers or oligomers. However, in a process where phase separation and percolation are coupled, reversible associations via specific interactions will enable the formation of an assortment of finite-sized, pre-percolation clusters (Li et al., 2012). Kar et al., demonstrated this for the protein FUS and other members of the FET family (Kar et al., 2022). They also showed that certain mutations can enhance cluster formation while weakening phase separation. In addition, mutations to the primary stickers that weaken cluster formation also weaken phase separation. Importantly, the clusters in question do not have fixed stoichiometries. Their size distributions are governed by sticker valence and the strengths of sticker interactions. These pre-percolation clusters form in deeply subsaturated solutions that are well below the average endogenous concentrations measured in cells (Hein et al., 2015). The upshot is that pre-percolation clusters, *and* condensates are likely to be of biological importance in different processes, and discerning their respective contributions needs to be a major focus.

One can also define a mutational strategy to identify the contributions to cluster formation versus phase separation. Formally, mutations to "pure spacers" can impact the driving forces for phase separation by changing c_{sat} without influencing the distributions of clusters in subsaturated solutions. Indeed, Kar et al., identified mutations in the RNA-binding domain of full-length FUS that achieve this functionality. Because the contributions of spacers are weaker than those of stickers, many spacer residues may have to be mutated to achieve decisive effects such as shifting c_{sat} without affecting clusters. Mutations to stickers will typically affect both c_{sat} and the distributions of clusters that form in subsaturated solutions. Therefore, mutations that change the extent of coupling between networking and density transitions can be used to shift the prevalence of clusters and condensates in cells. The functionality of such mutants will provide insight into the importance of clusters versus condensates for function.

PSCP AND THE STICKERS-AND-SPACERS FRAMEWORK HELP ADDRESS CHALLENGES THAT HAVE BEEN RAISED TO THE RELEVANCE OF PHASE SEPARATION IN CELLS

Recently, the relevance of phase separation in cell biology has been challenged on specific grounds (McSwiggen et al., 2019b). Four of the main challenges are as follows: (1) In live cells, certain systems do not show evidence for the existence of a saturation concentration, thus appearing to rule out phase separation as a relevant phenomenon. (2) Single particle tracking measurements appear to confound the expectation that condensates that form via phase separation should be defined by a phase boundary with a finite interfacial tension that hinders macromolecular transport across the boundary. (3) Because many cellular studies of phase separation utilize overexpression as a method for titrating macromolecular concentrations, it

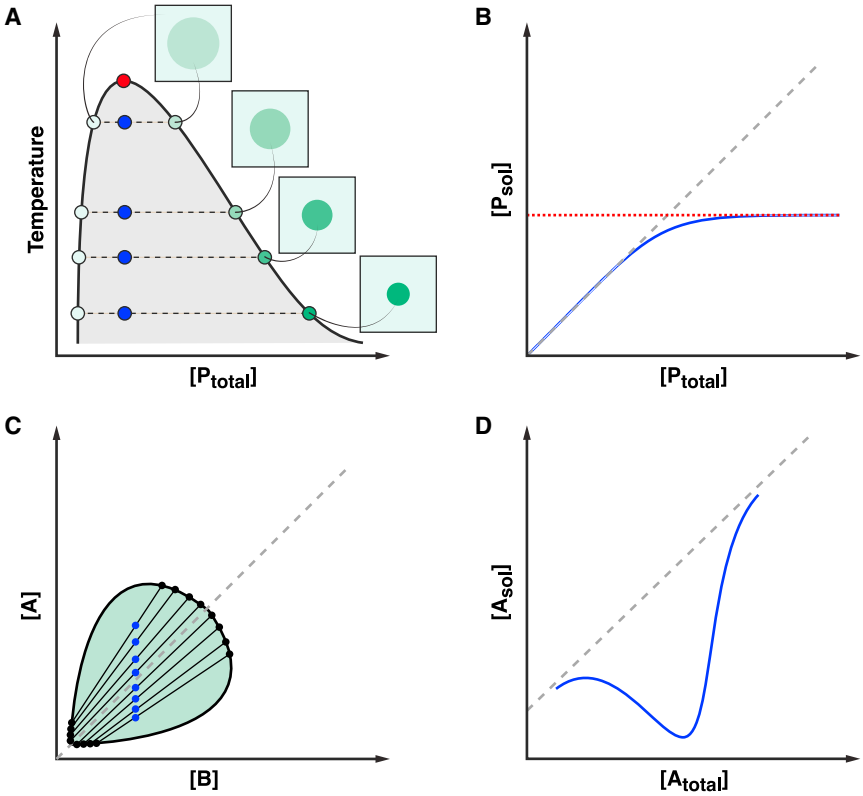

Figure 4. Dilute phase concentrations of macromolecules are determined by whether phase separation is driven by homotypic versus heterotypic interactions

(A) Shape of the phase boundary that results from a binary system of a macromolecule P and solvent. For a given total concentration P, the system separates into a dilute phase and a dense phase, and their densities are temperature dependent. The tie lines (broken lines) are horizontal because the temperatures are equal across the phase boundary.

(B) For increasing total concentration P, the soluble concentration increases up to the saturation concentration (indicated by the red broken line). Additional increase of the total protein concentration beyond the saturation concentration is incorporated into the growing dense phase volume fraction, and the dilute phase remains at c_{sat}.

(C) Shape of the phase boundary that results from a system where phase separation is driven purely by heterotypic interactions between macromolecules A and B. The black envelope is the phase boundary. The black lines are tie lines of constant chemical potentials, and they connect points corresponding to the joint concentrations of A and B in the coexisting dilute and dense phases. One can perform a thought experiment to ask how the concentration in the sol or dilute phase designated as $[A]_{sol}$ varies with the total concentration of [A] as the total concentration of B is held fixed. This would be a typical experiment to perform to test for the validity of phase separation by asking if $[A]_{sol}$ shows plateauing behavior above some value of [A].

(D) The dashed line is the 1:1 line between $[A]_{sol}$ and [A]. Within the two-phase regime, $[A]_{sol}$ (blue line) deviates from the 1:1 line, and it does not show the plateauing behavior expected of phase separation driven exclusively by homotypic interactions. Notice that although the bulk concentration of B is held fixed, the concentrations of B in the dilute versus dense phase are set by the tie line and not by the expression level of B. Therefore, along a tie line, the saturation of the dilute phase is set jointly by the concentrations of the A and B macromolecules, and the joint concentrations in the coexisting dilute phase change depending on the slopes of the tie lines.

follows that phase separation is irrelevant at endogenous expression levels. (4) Condensates that form via phase separation should undergo coarsening transitions, which are rarely observed *in vivo*, thereby challenging the relevance of phase separation in molecular and cell biology. Here, we address these challenges through the lens of PSCP.

Challenge 1: Certain systems do not show evidence for the existence of a saturation concentration

The discussion to this point has focused on the physics of binary mixtures comprising one type of macromolecule in a solvent. However, most biologically relevant mixtures are complex, featuring multiple different types of macromolecules and a solvent that comprises a mixture of water, salts, solutes, and metabolites. In the simplest generalization, namely a ternary mixture with two types of macromolecules in a complex solvent, the overall phase behavior is determined by the balance of solvent-mediated homotypic and heterotypic interactions. Here, homotypic interactions refer to the effective interactions between macromolecules of the same kind, whereas heterotypic interactions refer to the effective interactions between different kinds of molecules.

In a binary mixture of macromolecule and solvent, one can assess whether a condensate forms via phase separation by performing a concentration titration to determine if there is a threshold concentration c_{sat} above which the one-phase regime becomes saturated (Wang et al., 2018) (Figure 4). If a condensate arises via phase separation in a multicomponent system, does it automatically follow that the concentrations of all macromolecular components will remain fixed in the dilute phase? The answer is no. This is because the shapes of multidimensional phase diagrams are dictated by the extent to which homotypic interactions and heterotypic interactions work with one another to drive condensate formation.

The concentrations of each of the macromolecular constituents in the coexisting dilute phase will be determined by the shape of the multidimensional phase diagram and the slopes of tie lines. These are the lines of constant chemical potentials (Choi et al., 2019; Figure 4). Accordingly, concentrations in the coexisting dilute phase are governed by stoichiometric ratios of macromolecules that contribute to homotypic *and* heterotypic interactions (Guillén-Boixet et al., 2020; Pak et al., 2016; Riback et al., 2020; Yang et al., 2020). Therefore, the dilute phase is differently saturated by different stoichiometric ratios of the relevant, condensate-driving macromolecules (Riback et al., 2020). The complexity has been discerned for ternary systems (Banani et al., 2016; Choi et al., 2019; Deviri and Safran, 2020; Li et al., 2012). Furthermore, a recent study has suggested that the concept of a solubility product may serve as a useful proxy for assessing whether a multicomponent condensate arises via

phase separation (Chattaraj et al., 2021). Accordingly, along tie lines, the products of the dilute phase activities of macromolecular species that drive phase separation should be constant. If there are stoichiometric ratios of macromolecules that fit these criteria and give rise to condensates, then a reasonable inference is that phase separation is driving condensate formation. Importantly, if we know that a blend of homotypic and heterotypic interactions drives the formation of a condensate, it follows that the dilute phase is saturated by a convolution of macromolecular concentrations and not just the concentration of one type of macromolecule.

Challenge 2: Macromolecular transport should always be hindered across a phase boundary

For a probe molecule of radius R, the interface between the coexisting phases can set up a potential energy well (Münchow et al., 2008). The depth of this well, written as U_{min}, will depend on R, the contact angle with the interface, and the interfacial tension. The dwell time at the interface, designated as t_{dwell}, will be proportional to $\exp(-U_{min}/k_BT)$. Typical values for R and the interfacial tension are 10 nm and 10^{-4} N/m, respectively. Note that the measured interfacial tensions are 4–5 orders of magnitude lower than surface tensions at air-water interfaces (Atefi et al., 2014; Brangwynne et al., 2011; Feric et al., 2016). For such ultralow interfacial tensions, the well depth of the minimum at the interface that creates a finite dwell time will be on the order of thermal energy. In this scenario, the interface does not create a barrier for transport, and models that only account for partition coefficients, without giving special consideration to kinetic traps at the interface, show excellent agreement with measurements across interfaces (Münchow et al., 2008). The general conclusion from the physical literature is that an interface creates hindrances to molecular transport if and only if a back-to-back electric double layer forms at the interface (Hahn et al., 2011). Otherwise, to first order, the main determinants of transport across interfaces are partition coefficients and the intrinsic mobilities within the coexisting phases.

It is worth noting that most measurements suggest that the cytoplasm or nucleoplasm are viscoelastic (Bergeron-Sandoval et al., 2021; Berret, 2016; Erdel et al., 2015; Guigas et al., 2007; Xie et al., 2022) or even "poroelastic" materials (Hu et al., 2017). In addition, if a condensate forms via phase separation and it too is viscoelastic, then the intrinsic mobilities in the coexisting phases will be quite similar. In this scenario, the only determinant of transport across a phase boundary becomes the partition coefficient. Therefore, the ultralow interfacial tensions and the viscoelastic nature of coexisting phases significantly minimizes the extent to which molecular transport across a phase boundary will be hindered. These quantitative considerations must be accounted for (Kaur et al. (2021)) prior to setting up expectations regarding the timescales and mechanisms of molecular transport across phase boundaries.

Challenge 3: If the observation of condensates requires overexpression, then phase separation is irrelevant in cells

Almost all membraneless bodies that are postulated as being condensates that arise via phase separation are multicomponent entities encompassing multiple types and numbers of macromolecules (Decker and Parker, 2012; Yang et al., 2020). In such systems, the phase behavior is likely to be governed by a combination of homotypic and heterotypic interactions. For an n-component system, phase separation is described by an n-component density vector $[\rho_1, \ldots, \rho_n]$. Assessment of whether a condensate arises via phase separation requires a series of concentration titrations, whereby the concentrations of each of the n components and different combinations of these components are titrated. Without these concentration titrations, one cannot assess whether there exists a saturation of the dilute phase and if that is set by one or more components. Therefore, concentration titrations, achieved by altering expression levels or optogenetic manipulations, are imperative for assessments of whether phase separation is involved in the formation of condensates.

Next, we require a comparison of the endogenous levels to the measured phase boundaries. If the endogenous concentrations of the macromolecules that drive density transitions are lower than the measured or inferred threshold concentrations, then phase separation is likely not the operative mechanism for condensate formation. However, it is worth noting that endogenous levels are seldom fixed quantities. They change in response to stimuli, stresses, and cell states. Furthermore, recent studies have demonstrated that the concentrations of ligands, post-translational modifications, and the presence of surfaces can all lower the threshold concentrations for phase separation (Morin et al., 2022; Ruff et al., 2021a, 2021b). Therefore, the endogenous levels of a macromolecule, measured under generic conditions (Hein et al., 2015), provide limited information regarding the phase behaviors that are realizable in complex mixtures comprising an assortment of distinct multivalent macromolecules in different stoichiometric ratios. It is worth reiterating that *the overall saturation of the dilute phase* is governed by a single macromolecule if and only if homotypic interactions of this macromolecule are the drivers of phase separation. In the presence of a blend of homotypic and heterotypic interactions, the stoichiometric ratios of the relevant macromolecules will contribute jointly to setting the saturation threshold of the dilute phase through a convolution that will be system specific.

Macromolecules with sticker-and-spacer architectures that undergo PSCP will feature clusters of different sizes, shapes, internal structures, and compositions even in subsaturated solutions. For such systems, if the endogenous concentrations are in the subsaturated regime, then the relevant species are prepercolation clusters that form in subsaturated solutions. If the endogenous concentrations are in the supersaturated regime, then the system of interest is in the two-phase regime and the relevant species are both the condensate and the clusters that are present in the coexisting dilute phase. The presence of dynamical clusters, deemed to be inconsistent with condensates that form via phase separation, have been reported for specific systems in the context of transcriptional regulation (Chong et al., 2018; McSwiggen et al., 2019a). However, the multitude of clusters that can form for sticker-and-spacer systems undergoing PSCP provides a clear bridge between clusters in subsaturated solutions or coexisting dilute phases and

condensates in supersaturated solutions (Kar et al., 2022). What is required is a direct assessment of the distinct or shared functions of clusters that form in subsaturated solutions and condensates that form in supersaturated solutions

Challenge 4: Coarsening behavior is not observed in live cells

Based purely on thermodynamic considerations, one expects that the size distributions of dense phase droplets will change over time. Although many small condensates are present at the initiation of phase separation, the numbers of condensates will decrease and the sizes of those that persist will increase via coalescence (Berry et al., 2015; Lee et al., 2021). The final ground state is expected to be a single large condensate that coexists with a dilute phase. However, the observed size distributions of condensates in cells generally deviate from this expectation. Explanations for these deviations are multifold: From a purely passive standpoint, in the absence of sources or sinks, the interplay between the dynamics of making and breaking physical crosslinks versus the timescales for molecular transport can lead to valence saturating effects that hinder the growth of condensates (Ranganathan and Shakhnovich, 2020). A balancing of viscous and elastic forces can give rise to long-lived species of very similar sizes that do not fuse and coarsen over long periods of time (Dahiya et al., 2016). This arises directly from the viscoelastic nature of condensates. Active processes, such as protein production that is synchronized with protein degradation, can provide exquisite control over condensate sizes (Weber et al., 2019; Wurtz and Lee, 2018). In addition, the presence of cofactors that act either as Pickering agents (Folkmann et al., 2021) or surfactant-like polymer brushes (Cuylen et al., 2016) can act as emulsifiers that control the sizes of condensates within cells. In addition, cellular structures can prevent condensates from physically interacting as was shown in large oocytes, which often have a multitude of small nucleoli. The nuclear actin filament network prevents nucleoli from coalescing, and depolymerizing the network results in a single large nucleolus (Feric and Brangwynne, 2013). Other cellular structures and surfaces can be expected to have similar steric and/or other effects. Finally, block copolymeric systems, studied recently in the context of paraspeckles, can form micelles, which are exemplars of systems that undergo microphase separation that do not coarsen and instead form microphases or micellar phases with uniform size distributions (Yamazaki et al., 2021).

Other issues and clarifications

The nomenclature of liquid-liquid phase separation (LLPS) creates confusions because without any clarifications, this term can be taken to mean that the coexisting phases are simple, purely viscous liquids. However, although the dilute phase *in vitro* may be a simple, purely viscous Newtonian fluid, characterized by water-like viscosity, a uniform dielectric constant, and macroscopic surface tension, the coexisting dense phase is almost certainly a complex, viscoelastic material (Alshareedah et al., 2021b; Bergeron-Sandoval et al., 2021; Ghosh et al., 2021; Jawerth et al., 2020; Shen et al., 2020; Zhou, 2021). Viscoelastic materials are described by time-dependent stress-strain relationships, underlying network structures and a combination of viscous and elastic moduli. *In vivo*, the situation becomes more complex. As previously mentioned, the cytoplasm or nucleoplasm are viscoelastic materials (Bergeron-Sandoval et al., 2021; Berret, 2016; Erdel et al., 2015; Guigas et al., 2007; Xie et al., 2022). In addition, if a condensate that forms via phase separation is viscoelastic, then the process of phase separation cannot be referred to as being LLPS because the coexisting phases are not simple liquids. Instead, they are different types of viscoelastic network fluids or just viscoelastic materials. The relevant phase behavior is likely PSCP (Harmon et al., 2017; Li et al., 2012). Clearly, we must account for phase separation *plus* percolation, *plus* other transitions such as conversion to fibrillar solids (Kato et al., 2012; Murray et al., 2017), the formation of liquid crystalline phases (Rog et al., 2017; Scheff et al., 2020), or even active processes such as motility-induced phase separation (Cates and Tailleur, 2015). Accordingly, we propose the term phase separation++ or PS++ as a succinct alternative to capture the fact that phase separation can be coupled to a variety of other phase transitions *in vivo*. This implies that the time has come to drop "LL" as a prefix as it glosses over the complexities that have always been part of rigorous narratives on biological phase separation (Alshareedah et al., 2021b; Boeynaems et al., 2019; Choi et al., 2019, 2020a, 2020b; Feric et al., 2016; Harmon et al., 2017; Jawerth et al., 2020; Kar et al., 2022; Li et al., 2012; Roberts et al., 2018; Seim et al., 2022; Zhou, 2021).

The timescales for fluorescence recovery after photobleaching (FRAP) are typically used to assess if a condensate is liquid or solid like. It is worth noting that different network structures and dynamics are expected to result in different timescales of FRAP (Ma et al., 2021). For example, Boeynaems et al., show that even in simple protein-RNA mixtures, the recovery times for proteins are rapid and that they demonstrate near complete recovery after photobleaching (Boeynaems et al., 2019). Conversely, the RNA molecules are relatively immobile, and however, the complexes undergo phase separation defined by joint saturation concentrations, even showing the ability to fuse, albeit slowly, into well-rounded structures (Boeynaems et al., 2019). These observations, gleaned from measurements other than FRAP, clearly demonstrate the weakness of FRAP as a measure of anything other than molecular mobilities. Comparative measurements of FRAP are useful, but they are not measures of material properties. Furthermore, whether condensate components drive phase separation or are being recruited to condensates is also expected to alter how strong their networking is in condensates and hence how quickly their fluorescence recovers after bleaching. Differences are thus likely to be a feature of programmable biochemistry rather than the lack of prescribed quantitative criteria for defining fast FRAP (McSwiggen et al., 2019b). Given that fast FRAP can be the result of structures other than liquids, e.g., spongy structures that transiently bind the recovering species, FRAP cannot be used as a method to decide if a condensate has liquid-like or other properties. This point that has been made previously (Alberti et al., 2019; Taylor et al., 2019).

Finally, considerable attention has focused on intrinsically disordered regions (IDRs) as drivers of phase separation. As recently discussed, the inadvertent leap, whereby the presence of an IDR is taken to imply that the protein of interest will drive phase separation via homotypic interactions is problematic on

many levels (Martin and Holehouse, 2020). First, it ignores over a decade of work demonstrating that IDRs come in a wide range of sequence flavors (Das et al., 2015; Martin and Holehouse, 2020), and many IDRs encode negative values of the Flory χ parameter, and hence, an inability to drive phase separation through homotypic interactions (Harmon et al., 2017, 2018). Only if IDRs encode sticker valence that is neither too low nor too high and their solubility (or the solubility of their complexes) is low enough, can we expect them to drive phase separation. The corollary is that many IDRs are highly soluble, and they mediate functions independent of driving phase separation (Cohan et al., 2020; Forman-Kay and Mittag, 2013; Martin and Holehouse, 2020; Sherry et al., 2017; Wright and Dyson, 2015).

BIOLOGICAL PROCESSES THAT ARE LIKELY MEDIATED BY PSCP

Prior to concluding this perspective, we highlight examples in biology where a strong case can be made for complex mixtures undergoing PSCP. The *nucleolus* can be observed by light microscopy due to its sheer size and density. Nucleoli assemble around the gene clusters encoding ribosomal RNA (rRNA) and require outflux of rRNA and influx of ribosomal proteins for their formation. This counter-flux gives rise to a distinctive layered structure, which is critical for the stepwise assembly of ribosome subunits. The ability of the different layers to fuse and coalesce independently from each other points to interfacial tension between them and is evidence that nucleoli are in fact multilayered condensates (Feric et al., 2016). A coexistence line was clearly demonstrated for the granular component of nucleoli by manipulation of cell size and protein concentration (Weber and Brangwynne, 2015). rRNA transcription shapes the assembly of nucleoli in a manner consistent with phase separation that is modulated by active processes in cells (Berry et al., 2015) and in living drosophila embryos (Falahati et al., 2016). Although the exact mechanisms of how PSCP facilitates ribosome assembly still needs to be characterized, nucleoli appear to assemble by a combination of passive and active PSCP, where the process likely involves some form of complex coacervation (Pak et al., 2016).

Stress granules are cytoplasmic structures that form under stress conditions that disassemble polysomes. Recent work has shown that they are formed via phase separation of naked RNA (released from disassembling polysomes) and RNA-binding proteins (Guillén-Boixet et al., 2020; Sanders et al., 2020; Yang et al., 2020). G3BP is the central node in the stress granule interaction network when stress granules form in response to arsenite as a stressor. The domain contributions to the *in vitro* phase behavior of G3BP with RNA recapitulate the domain contributions to stress granule assembly in cells, providing strong support that phase separation is the mechanism underlying stress granule assembly (Guillén-Boixet et al., 2020; Sanders et al., 2020; Yang et al., 2020). Furthermore, highly accurate stress granule mimics can be assembled in lysates; the addition of small amounts of G3BP to lysates nucleates condensates that faithfully recapitulate the composition of stress granules (Freibaum et al., 2021). The tenets of PSCP apply demonstrably to stress granules induced by arsenite stress, and G3BP is the key node in the assembly of this complex viscoelastic network fluid.

Other stress-related condensates

The contribution of certain proteins to the formation of stress-related condensates has been characterized and provides powerful evidence that they are formed by phase separation. Poly(A)-binding protein 1 (Pab1) is a key factor in the assembly of yeast heat-shock granules. The driving force for its assembly into condensates is tunable by the amino acid composition of its proline-rich IDR, although the IDR is not the driver of phase separation. Yeast cells expressing mutant Pab1 variants with decreased driving force for phase separation show a growth defect under heat stress conditions, and this fitness effect was titratable in accordance with the *in vitro* phase behavior of Pab1 variants (Riback et al., 2017). These data strongly suggest that Pab1 is a major factor in mediating the formation of phase separated, adaptive heat-shock granules in yeast. Similarly, the phase behaviors of Pub1 and Sup35 are direct predictors of cellular fitness in yeast upon energy depletion (Alberti et al., 2018; Kroschwald et al., 2018).

Regulation of local translation and signaling by phase separation

Recent evidence implicates phase separation as the main factor that regulates local translation in neurons, which is important for memory formation. The two RNA-binding proteins FMRP and CAPRIN1 can form mixed or multiphasic condensates and recruit or exclude RNA depending on the post-translational modification status of the two proteins. This tunes the relative activities of RNA deadenylation and translation by several orders of magnitude when reconstituted *in vitro* (Kim et al., 2019). Multicomponent, PTM-regulated PSCP is therefore a plausible mechanism underlying the known cellular regulation of RNA degradation versus translation by RNA-binding proteins.

In several systems, strong evidence from mutagenesis and manipulation of protein levels points toward the importance of PSCP for enhancing cell signaling. Phase separation has been invoked as the mechanism underlying noise filtering downstream of membrane receptors, specifically of the T cell receptor and Nephrin (Case et al., 2019; Huang et al., 2019). Decisive signaling steps by molecules such as SOS and Arp2/3 downstream of the receptors have slow activation processes, which only proceed if the molecules are trapped in long-lived clusters mediated by multivalent interactions. The ability to access percolated network structure within condensates may be the effective benefit of phase separation in these systems.

Finally, the phase behaviors of oncogenic fusion proteins such as Nup98-HOXA1 have been implicated as the underlying mechanism for wholesale oncogenic changes to the transcriptional program and chromatin structure of cells expressing these factors (Ahn et al., 2021; Chandra et al., 2022). Weakening the drive for phase separation reduces the oncogenic potential of these proteins. The IDR that contributes a major part of the driving force for PSCP can be replaced by other IDRs of different phase separating potentials, thus providing tunability and interoperability. In principle, the observed effects might be attributable to clusters in dilute phases instead of condensates, and this will need to be tested in the future. Distinct lines of evidence exist for active roles of clusters and repressive roles for condensates (Chong et al., 2022; Powers et al., 2019). Alternatively,

condensates are thought to be the relevant entities that regulate gene expression at super enhancers (Sabari et al., 2018), partly through active remodeling of condensates by newly synthesized transcripts (Henninger et al., 2021). It is likely that the active versus repressive functions of pre-percolation, finite-sized clusters in coexisting sol versus viscoelastic condensates will be context dependent, and this context dependence will require scrutiny.

CONCLUSIONS

The case for phase separation or more precisely PSCP/PS++ has been made in a variety of settings including the organization of genomes (Gibson et al., 2019; Larson et al., 2017), the regulation of transcription and translation (Boija et al., 2018; Guo et al., 2019; Lu et al., 2018), buffering of noise in transcription (Klosin et al., 2020), the appropriate reception of external signals and their transduction into intracellular responses (Case et al., 2019; Huang et al., 2019; Su et al., 2016), shaping of membranes (Agudo-Canalejo et al., 2021; Bergeron-Sandoval et al., 2021; Yuan et al., 2021), and stress responses (Alberti et al., 2018; Kroschwald et al., 2018; Riback et al., 2017; Yang et al., 2020). At this juncture, the pressing need is to go beyond simple systems, which are extremely helpful for identifying key rules, and develop new approaches for describing phase transitions such as PSCP and their functional consequences in multicomponent systems. This will require a deeper appreciation of physical concepts that describe the interplay between equilibrium and non-equilibrium phenomena. It might even require the development of new physics to describe what are essentially non-macroscopic systems. A blended approach that embraces biochemical reconstitutions, cellular studies, computational advances, and predictive theories will need to be brought to bear.

PS++ provides a route for incorporating spatial and temporal organization into systems biology approaches for modeling integrated cue-signal-response triads of individual cells. Within a cell, the concept of a condensate arising from PS++ opens the door to describing functions on the mesoscale—the in-between scale that lies above the molecular scale and below the cellular scale. In this context, it is worth drawing attention to the perspective written by the physicist Phillip Anderson, which was entitled *More is different—broken symmetry and the nature of hierarchical structure in science* (Anderson, 1972). Adapting his ideas, we note that the "constructionist hypothesis" of building up higher-order assemblies, one set of specific interactions at a time, rests on the presumption that structures, dynamics, and functions on the mesoscale are, for the most part, an exact sum of what we can measure and record at the molecular scale. The PS++ view being advanced through a blend of physics, chemistry, biophysics, biochemistry, and cell biology rests on an emergent hypothesis as opposed to a constructionist one. Accordingly, bottom-up biochemical reconstitutions and top-down measurements in live cells need to go hand in hand to identify the length scales where new properties and complexities emerge at the mesoscale.

In closing, we note that prior to the revolutions of molecular and structural biology, the fields of biomacromolecules and synthetic polymers tended to feed off one another. These fields diverged a few decades ago, but now, with renewed focus on intracellular phase transitions, we are realizing that there is much to be learned from the parallel disciplines that occupy the domain of polymer sciences. Specificity, a defining hallmark of biochemistry, can be readily engineered into synthetic polymers. In addition, the realization that disorder and conformational heterogeneity can engender new forms of specificity lead us to map biomacromolecules onto synthetic systems such as associative polymers and patchy colloids (Bianchi, 2019; Sanders et al., 2020). Accordingly, advances made in mechanistic biochemistry, structural biology, and biophysics when combined with advances in polymer sciences offer constructive ways to help us understand how emergent properties on the mesoscale arise and are regulated to influence cellular functions.

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DECLARATION OF INTERESTS

T.M. is a consultant for Faze Therapeutics. She is also a member of the editorial advisory board of *Molecular Cell*. R.V.P. is a member of the Scientific Advisory Board of Dewpoint Therapeutics Inc. The authors declare that these affiliations have not influenced or compromised the perspective provided here.

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